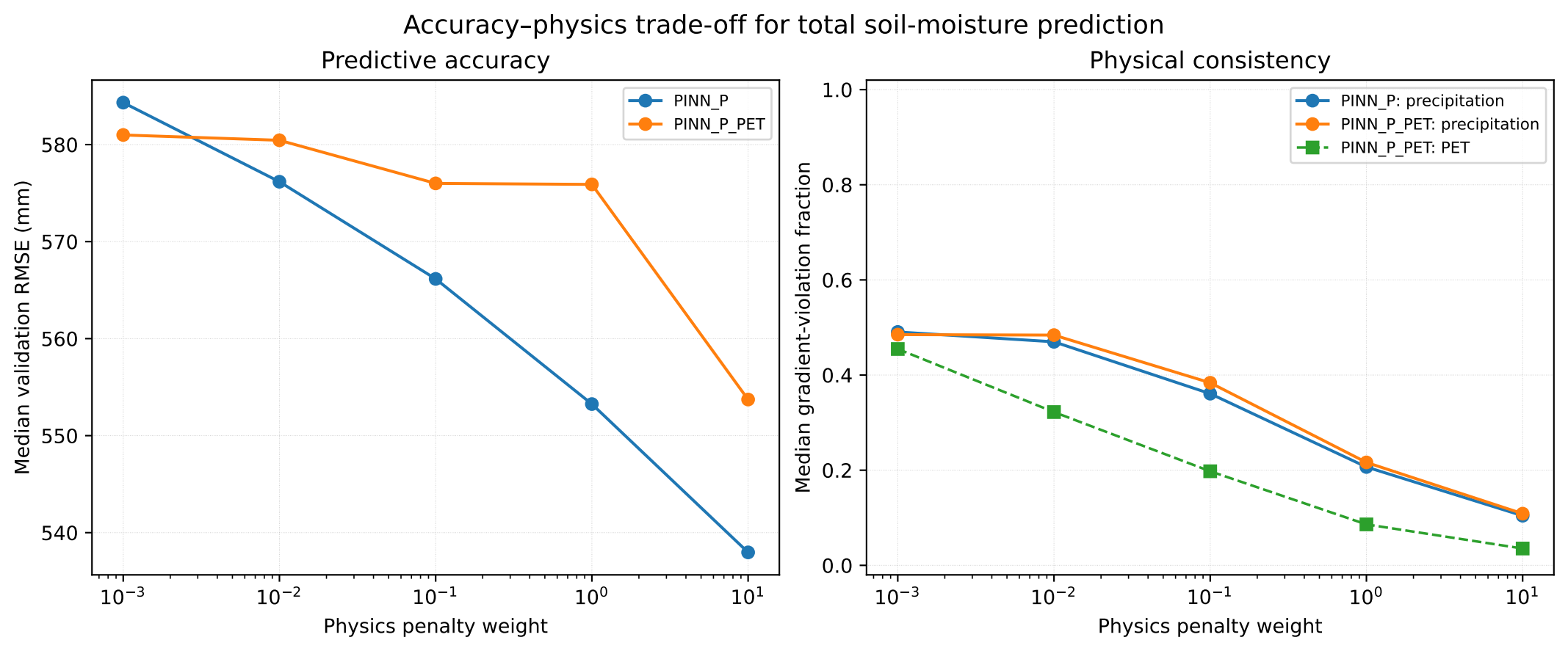
Accuracy–physics trade-off for total soil-moisture prediction
Predictive accuracy
Physical consistency
Median validation RMSE (mm)
Median gradient-violation fraction
Physics penalty weight
Physics penalty weight
PINN_P
PINN_P_PET
PINN_P: precipitation
PINN_P_PET: precipitation
PINN_P_PET: PET
580
570
560
550
540
1.0
0.8
0.6
0.4
0.2
0.0
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}